

Liquid stability, LLE and VLLE

After midterm · Session 3 of 6 · 180 minutes

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Session outcome

Use global Gibbs-energy support to assess candidate liquid phases.

Before class: Review gmix, local curvature and the distinction between stationary and globally stable states.

Learning path and all labs

The 180-minute class

IN CLASS	MINUTES
Recall and prediction	10
Concepts and derivation	45
Worked example	30
Break	10
Instructor lab demonstration	25
Guided student exploration	35
Discussion and interpretation	15
Exit question and independent task	10

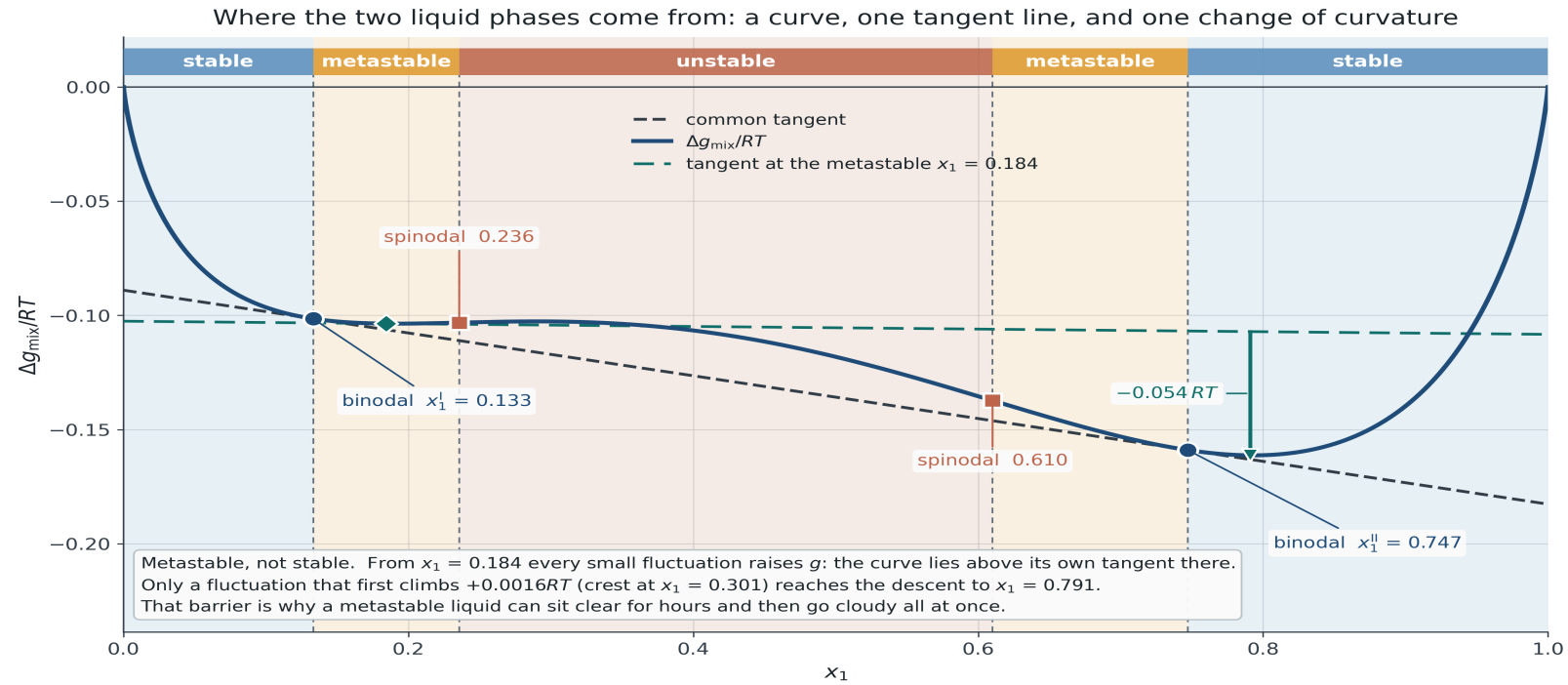
Phase equilibrium includes stability

Equality of chemical potentials identifies stationary candidate states. Stable equilibrium requires the lowest admissible total Gibbs energy.

$$g_{mix}/RT = \sum_i x_i \ln x_i + g^E/(RT)$$

A split can lower G even when a homogeneous liquid is locally stable.

Binodal and spinodal



ILLUSTRATIVE PARAMETERS: three-suffix Margules with $A_{12} = 2.6$, $A_{21} = 2$, chosen to place a partially miscible binary on one screen. They are not fitted to any measured mixture. Every number printed above is computed from that one G^E : the binodal by `vlekit.flash.miscibility_gap`, the spinodal by `vlekit.flash.spinodal`, the barrier and the descent by differencing the curve against its own tangent.

Module 5 reference illustration: coexistence and local instability are different boundaries.

The common-tangent conditions

$$g'(x^\alpha) = g'(x^\beta) = \frac{g(x^\beta) - g(x^\alpha)}{x^\beta - x^\alpha}$$

The tangent must not lie above $g(x)$ anywhere in the admitted composition interval.

A trivial root $x^\alpha = x^\beta$ does not establish two distinct liquid phases.

Tangent-plane distance

For a trial composition w relative to reference z ,

$$TPD(w; z) = g(w) - g(z) - g'(z)(w - z)$$

A negative value identifies a Gibbs-energy-lowering perturbation. Local curvature examines only the neighborhood of z .

Lab 04 evaluates the binary model; it is not a general multicomponent TPD package.

A lever-rule calculation

Synthetic tie line: $x_1^\alpha=0.2$, $x_1^\beta=0.8$. Feed $z_1=0.5$.

$$f_\beta = \frac{z_1 - x_1^\alpha}{x_1^\beta - x_1^\alpha} = 0.5$$

For $z_1=0.35$, $f^\beta=0.25$. At the same coexistence conditions, the phase amounts change but the tie-line endpoints do not.

Model shape and liquid splitting

Positive activity deviations alone do not guarantee LLE. The mixing Gibbs-energy shape must support a split.

The labs include Margules and NRTL options with different shapes. Standard Wilson behavior is discussed in the reference deck, not implemented as an LLE solver here.

LLE fitting uses different evidence

Tie-line data constrain the coexistence compositions. Fitting must reproduce distinct phases and a supported common tangent.

A small tie-line error at one T does not establish a unique parameter set or a temperature law.

Lab 05 retains the input observations and checks stability of the fitted result.

Class demonstration • Labs 04 and 05

Stability and LLE • LLE fitting

1. Compare convex and nonconvex mixing Gibbs curves.
2. Inspect global support and the phase fractions.
3. Fit a tie-line example at its stated temperature and inspect the stability check.

Guided exploration • 35 minutes

Choose one state with a supported liquid split.

Predict the effect of changing z while holding T and model fixed. Record phase compositions and amounts, then check the lever rule.

Use a fitting example to state one parameter uncertainty that a small residual cannot resolve.

Vapor–liquid–liquid coexistence

$$\hat{f}_i^V = \hat{f}_i^{L\alpha} = \hat{f}_i^{L\beta}$$

For a binary nonreacting system with three phases, the phase rule gives one intensive degree of freedom before imposing T or P.

Phase compositions and phase amounts are different unknowns. Specifying T and P does not always uniquely determine all amounts.

Three phase amounts can form a family

Given three fixed binary phase compositions,

$$f_{\alpha} + f_{\beta} + f_V = 1$$

$$z_1 = f_{\alpha} x_1^{\alpha} + f_{\beta} x_1^{\beta} + f_V y_1$$

There are two independent constraints on three fractions. Nonnegativity bounds the feasible family.
Another independent extensive constraint may select a member.

VLE extension

Open Lab 06

Compare the supported liquid and vapor states. Move within the admissible phase-fraction family and verify the overall balance.

Identify a state where the vapor fraction is constrained by nonnegativity. Explain why equal fugacities alone cannot determine every fraction.

Where the model can be reused

A fitted liquid model needs a matching component order, parameter convention and temperature before reuse.

VLE also requires pure-component vapor pressures and a vapor model. LLE experimental pressure is not a vapor-pressure correlation.

A new phase calculation brings new assumptions and validation needs.

Exit question

A solver reports two liquid compositions with equal chemical potentials and a tiny residual.

What checks distinguish a real stable split from a trivial or metastable result?

Which quantities change when only the overall composition changes?

Independent practice • suggested 60–90 minutes

A common-tangent result, one lever-rule check and one limitation of the fitted parameters.

Retain the calculator export, your worksheet, a comparison plot/table and one independent check.

State an assumption that limits your conclusion.

Use the core labs on the [learning path](#). Optional extensions are additional work.

References and further study

[Module reference deck](#) · [Lab sources and equations](#)

Derivations and original figure references remain in the corresponding module deck. Each lab records its implemented equations and assumptions.

Synthetic worked examples illustrate calculations; they are not evidence of real-system accuracy.